TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
Semester II 2026

Simplex Method Project

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The Simplex Algorithm

The Simplex Algorithm is a mathematical method used to solve linear programming problems. It was developed by George Dantzig in 1947 and has since become one of the most widely used optimization techniques in various fields, including operations research, economics, and engineering.

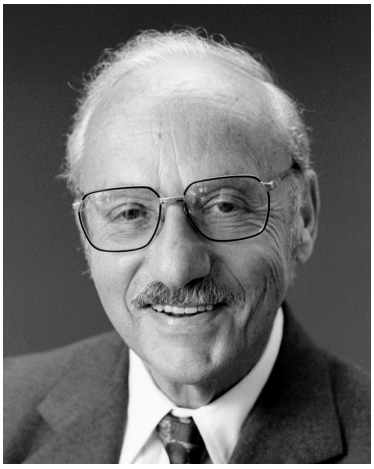


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

Original Problem

Description

- Problem Name: Some name
- Amount of decision variables: 8
- Amount of restrictions: 6
- Total Amount of variables: 15

Equations

$$
\begin{aligned}
-71.000000x_1 + -59.000000x_2 + 45.000000x_3 + 89.000000x_4 + 39.000000x_5 + 25.000000x_6 + -8.000000x_7 \\
44.000000x_1 + -50.000000x_2 + -96.000000x_3 + 52.000000x_4 + -50.000000x_5 + 62.000000x_6 + 54.000000x_7 \\
29.000000x_1 + 22.000000x_2 + -76.000000x_3 + -45.000000x_4 + -21.000000x_5 + 95.000000x_6 + -55.000000x_7 \\
68.000000x_1 + 23.000000x_2 + 31.000000x_3 + -85.000000x_4 + -6.000000x_5 + 14.000000x_6 + 2.000000x_7 \\
69.000000x_1 + 21.000000x_2 + 87.000000x_3 + -52.000000x_4 + -86.000000x_5 + -60.000000x_6 + 87.000000x_7
\end{aligned}
$$

Initial Table

Intermediate Tables

Final Table

Solution